

Can the best grower fail the brief?

Learning intention

Apply plant biology to unfamiliar growth data and justify a plan using evidence and its limitations.

Mission: recommend a growing plan with evidence.

A water budget arrives after the first decision.

Link plants with evidence

R1. How can photosynthesis contribute to biomass?

R2. Is water supplied the same as water lost?

R3. Does a group mean describe every individual?

I check what actually changed

A controlled comparison needs a fair pair

A: low light
Water supplied 0.30 L

B: high light
Water supplied 0.60 L

C: low light
Water supplied 0.15 L

D: high light
Water supplied 0.15 L

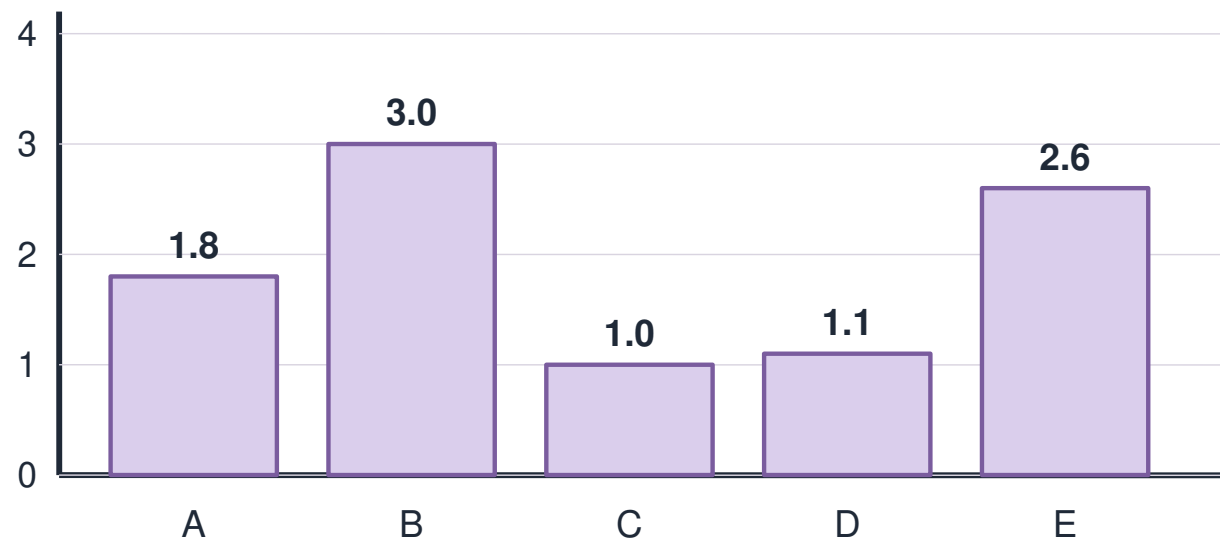
For a light test, compare C and D: same supplied water.

The table shows light and water supplied. A versus B changes both. C versus D keeps supplied water equal.

Read the outcome and its limits

Constructed results: growth is not the only rule

Mean dry-biomass gain(g/plant) over 14 days

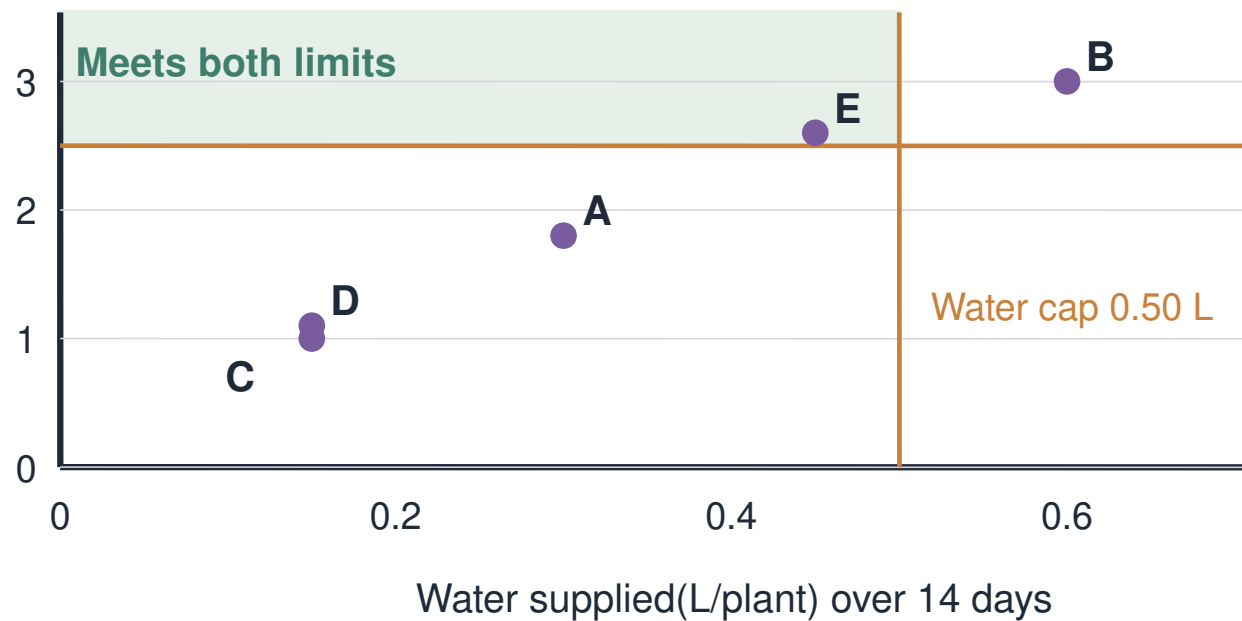


The outcome is mean dry-biomass gain. Dry mass reduces the effect of differing water content. A mean still hides individual variation.

I use two tests before recommending

Two requirements make a decision region

Mean dry-biomass gain(g/plant) over 14 days



Target: mean gain at least 2.5 g/plant. Budget: water supplied at most 0.50 L/plant. A plan must pass both.

Round 1: choose, then test the brief

W1. Which A–D plan has the greatest mean gain? Apply both targets. Does any A–D plan pass?

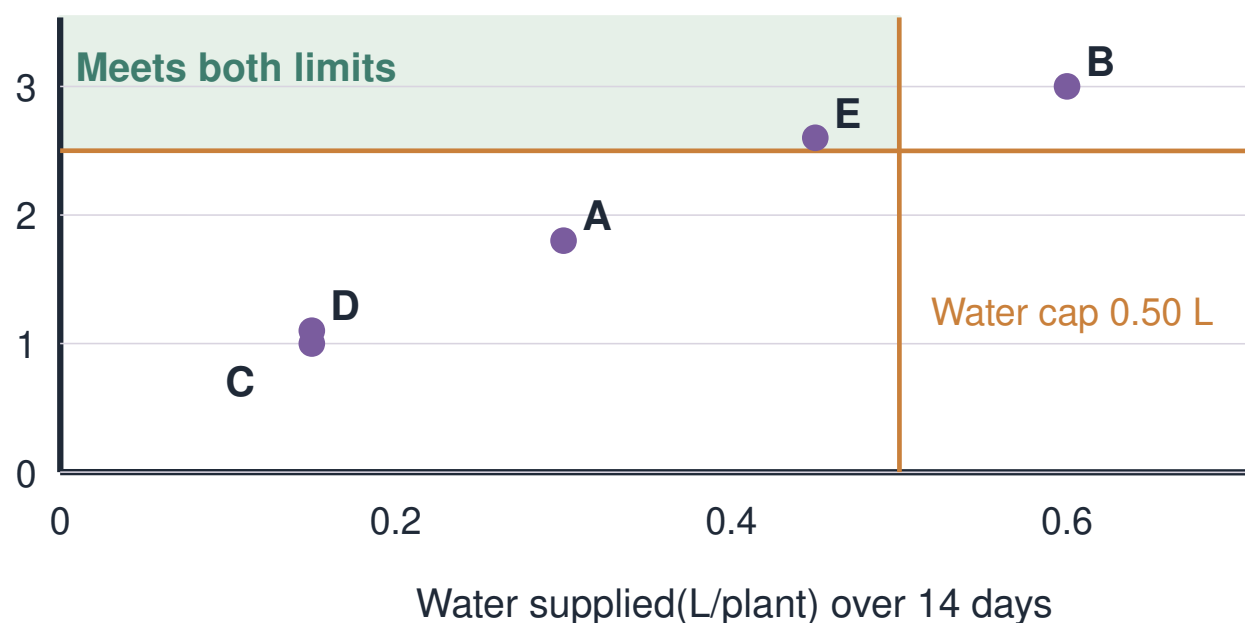
Plan	Light	Water supplied(L)	Mean gain(g)
A	Low	0.30	1.8
B	High	0.60	3.0
C	Low	0.15	1.0
D	High	0.15	1.1

Constructed 14-day data;5 plants per matched start/end group.

Round 2: new evidence changes the options

Two requirements make a decision region

Mean dry-biomass gain(g/plant) over 14 days



W2. Plan E: moderate light, 0.45 L supplied, 2.6 g mean gain. Does it meet both targets? Give both margins.

Round 3: keep the recommendation honest

W3. “E guarantees every plant succeeds.”

Improve the recommendation and request one check.

Which claim survives both checks?

Choose the claim with correct evidence and scope.

- A** B must win because 3.0 g is the largest mean.
- B** E meets both mean-based criteria in the supplied trial.
- C** A mean of 2.6 g guarantees every plant gains 2.6 g.
- D** A/B isolates light because both are plants.

Write a decision that can be checked

Choose your response route.

State a plan, evidence, mechanism and limit.

Make the recommendation useful

Check both conditions before choosing.

Link plant biology with evidence without claiming certainty.

Audience challenge: what could change your mind?

L1. Share your recommendation.

Audience asks for one piece of evidence that could overturn it.

A scientific decision remains testable

“Recommend E for a repeat pilot” is a claim with a next step.

A limitation makes the decision more useful.

Use the unfamiliar evidence

E1. Which A–D pair changes light with the same water total?

E2. Why does B fail the budget brief?

E3. Why is E a provisional recommendation?

Next useful step

Fair pair • two constraints • mechanism • uncertainty.